

Future Risks of Advanced AI Systems and Practical Strategies for Safer and More Reliable Artificial Intelligence

Research Proposal for Prospective Master's Supervision

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Abstract

Advanced artificial intelligence systems can create substantial benefits, but their growing capabilities and wider deployment also raise concerns about reliability, harmful misuse, privacy, bias, security, and the adequacy of human oversight. This proposal examines future AI risks and practical approaches for developing safer and more reliable AI systems. The study will use a structured qualitative literature review of peer reviewed research, technical reports, and international AI governance frameworks. The selected literature will be analysed thematically to identify major risk categories, assess proposed safeguards, and compare their limitations. The study will focus on measures that can be applied in practice, including model evaluation, red teaming, monitoring, human oversight, transparency, access controls, incident reporting, and lifecycle risk management. The expected outcome is an evidence-based framework that connects AI risk categories with practical technical, organizational, and governance responses. The study does not attempt to predict one certain AI future. Instead, it aims to clarify risks supported by current evidence and identify reasonable actions that developers, organizations, and policymakers can take under uncertainty.

1 Introduction

Artificial intelligence is increasingly used in education, software development, health, finance, public services, and creative work. Recent general-purpose and generative AI systems can produce text, images, code, recommendations, and decisions at a scale that makes their benefits and harms more consequential. Their rapid progress has strengthened interest in a central question: how can society obtain the benefits of advanced AI while reducing foreseeable harm?

The issue is not limited to hypothetical future scenarios. Current systems can provide false or unsupported information, reproduce social bias, disclose sensitive information, be manipulated by adversaries, or be misused to create harmful content. As systems become more capable, connected to tools, and deployed in high impact settings, risk management must address both present failures and future uncertainties. The International AI Safety Report groups concerns around malicious use, malfunctions, and systemic risks, while emphasizing that the future remains uncertain (International AI Safety Report, 2026).

This proposal therefore takes a balanced approach. It treats AI risk neither as a reason to reject technological progress nor as a problem that can be solved through one technical intervention. Safer AI requires a combination of robust technical design, careful evaluation, human oversight, organizational accountability, and governance across the AI lifecycle (NIST, 2024; OECD, 2024).

2 Background

Research on language models and other foundation models has documented several pathways through which harm can occur. These include misinformation, discrimination, privacy loss, unsafe or unreliable outputs, security vulnerabilities, and harmful use by malicious actors. Weidinger et al. (2022) provide a widely used taxonomy of language-model risks and show that harms can arise from both the model itself and the way it is deployed in social settings. Bender et al. (2021) also draw attention to concerns related to data, environmental cost, representational harms, and the tendency to treat fluent outputs as reliable knowledge.

Although many safety practices have been proposed, there is no single, practical synthesis that explains which safeguards respond to which risks and what their limitations are. Developers and institutions may know that AI should be trustworthy, but they still need actionable guidance: which risks should be assessed, how should systems be tested, when should a human review be required, and what should happen after deployment?

2.1 Problem Statement

The problem addressed by this study is the absence of a concise and practically useful synthesis that links future AI risks with safeguards that developers and organizations can apply. Existing work is often specialized: it may examine a single technical failure, a broad governance principle, or one type of AI system. This makes it difficult to decide which risks deserve attention, which interventions are suitable, and what limitations remain. The study will address this problem by translating the evidence into a structured framework appropriate for technical and organizational decision-making.

Three evidence based categories of AI risk

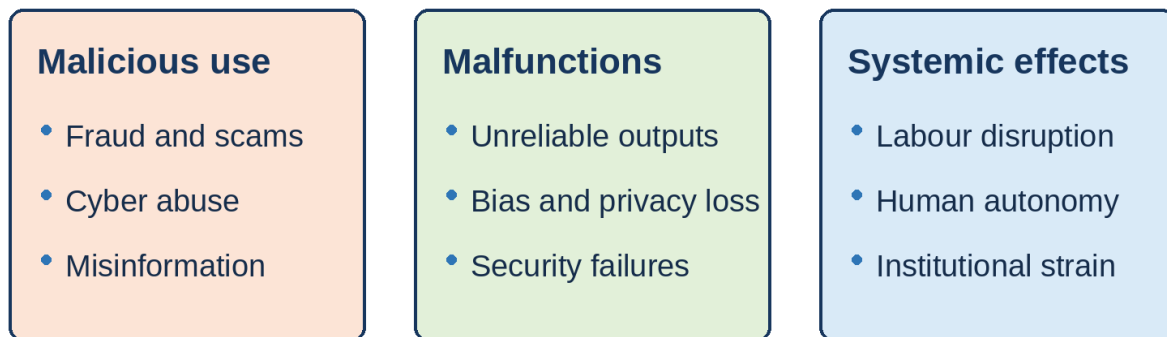


Figure 1. High-level AI risk landscape used in this study. Source: developed by the author from International AI Safety Report (2026).

2.2 Purpose of the Study

The purpose of this study is to bring together credible research and institutional guidance into one clear account of the risks associated with advanced AI and the safeguards that can realistically be used. It will give equal attention to current evidence and future uncertainty. The proposal is intended as a focused starting point for discussion with prospective faculty and will be refined under academic guidance to address a supervisor's research priorities.

2.3 Rationale for the Study

AI safety is often discussed either in highly technical language or through broad public warnings. This makes it difficult for a new researcher, developer, or institution to decide which risks are immediate, which are uncertain, and which safeguards are realistic. A focused literature review is needed to connect these discussions. The study is timely because generative and general-purpose AI systems are being adopted faster than many organizations can develop dependable evaluation, monitoring, and accountability practices.

3 Aim Objectives and Research Questions

3.1 Aim

The aim of this study is to examine major future risks associated with advanced AI systems and to identify practical approaches for building safer and more reliable AI systems.

3.2 Objectives

- Identify the main risk categories discussed in current AI safety literature and institutional frameworks.
- Examine technical, organizational, and governance approaches proposed to reduce these risks.

- Compare the strengths and limitations of major safety approaches.
- Develop a practical framework linking AI risks with appropriate safeguards.

3.3 Research Questions

- What current and future risks are most frequently associated with advanced AI systems?
- What practical technical, organizational, and governance approaches are proposed to improve AI safety and reliability?
- What limitations remain in existing approaches to AI risk reduction?
- How can identified risks and safeguards be organized into a practical framework for responsible AI development and deployment?

3.4 Hypothesis

A formal hypothesis is not proposed because this is a qualitative literature-review study rather than a quantitative or experimental study. The research will answer the stated questions by synthesizing evidence. If the study is later extended into an experiment, testable hypotheses can be developed from the proposed framework, for example by comparing the reliability of AI outputs before and after a specified safety intervention.

3.5 Key Terms and Study Boundaries

In this proposal, advanced AI refers to highly capable general-purpose or generative systems, including large language models and related foundation-model applications. AI safety refers to the prevention or reduction of harmful system behaviour and harmful use. AI reliability refers to the degree to which a system produces dependable, appropriately calibrated, and contextually suitable outputs. The study considers technical and sociotechnical risks; it does not conduct experiments on a particular model, estimate the probability of catastrophic outcomes, or make claims about a single inevitable future.

4 Literature Review

4.1 Risk Categories

The literature indicates that AI risk is multi-dimensional. Reliability risks include hallucinations, poor calibration, inconsistent behaviour, and failures in unfamiliar situations. Fairness and transparency concerns include biased outcomes and insufficient explanation of consequential decisions. Security risks include prompt injection, adversarial manipulation, data poisoning, and unauthorized access. Misuse risks include fraud, misinformation, cyber abuse, and the harmful automation of tasks. More advanced systems also raise questions about autonomy, monitoring, and whether human operators can retain meaningful control.

Hendrycks et al. (2021) organize machine-learning safety work into robustness, monitoring, alignment, and systemic safety. This classification is useful because it prevents the study from treating safety as only a content-moderation problem. Similarly, recent surveys of LLM safety discuss alignment, adversarial robustness, misuse, and autonomous-system risks as linked but distinct areas of research (Chua et al., 2024).

4.2 Practical Safety Approaches

Technical approaches include pre-deployment evaluation, red-teaming, adversarial testing, uncertainty estimation, retrieval from trusted sources, output verification, access control, and continuous monitoring. These methods aim to identify unsafe behaviour before and after a system is released. Organizational measures include clear accountability, documentation, human review for high-impact use, user reporting channels, and

incident-response processes. Governance measures include risk assessments, audit mechanisms, transparency obligations, and procedures to override, repair, or discontinue unsafe systems.

NIST (2024) recommends a lifecycle approach for generative AI, using governance, risk mapping, measurement, and management. UNESCO (2021) adds a human-rights and ethics perspective, emphasizing human oversight, fairness, privacy, and accountability. Together, these sources show that reliable AI is not only a model-quality issue; it is a sociotechnical responsibility involving people, institutions, and technical systems.

4.3 Review of Selected Studies

The selected literature gives this proposal a balanced foundation. The studies below were chosen because they are either peer-reviewed academic work or publications from recognized international institutions. They are used as evidence, not as predictions of a single inevitable AI future.

Source	Type	Main contribution	Use in this proposal
Weidinger et al. 2022	Peer reviewed ACM paper	Taxonomy of ethical and social risks from language models.	Risk categories and deployment context.
Hendrycks et al. 2021	Research paper	Frames safety around robustness, monitoring, alignment, and systemic safety.	Technical safety concepts.
NIST 2024	Official framework	Practical guidance for managing generative AI risk across the lifecycle.	Method and safeguards.
International AI Safety Report 2026	International scientific report	Synthesizes evidence on misuse, malfunctions, systemic risks, and risk management.	Future-risk context and uncertainty.
Gyevnar and Kasirzadeh 2025	Peer reviewed Nature article	Examines how AI-safety risks can be made understandable beyond specialist communities.	Accessible framing and significance.

4.4 Research Gap

Existing research provides valuable but fragmented guidance. Some studies focus on particular technical failures, such as robustness, alignment, or security, while governance frameworks describe broad principles for responsible AI. Reviews also commonly concentrate on one model type, especially large language models, or discuss risks without linking each risk to a practical safeguard and its limitations. There is therefore a gap for a concise, evidence-based synthesis that brings future AI risks, present reliability problems, technical controls, organizational practices, and governance measures together in one framework. This study addresses that gap by mapping the major risk categories to layered and realistic safeguards, while making the remaining uncertainty and limitations explicit.

5 Conceptual Framework

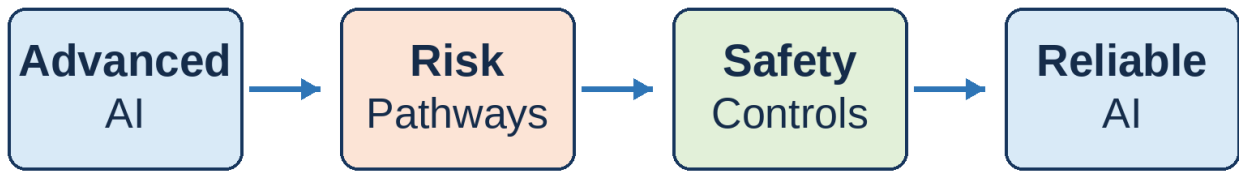


Figure 2. Conceptual pathway used in this proposed study. Source: developed by the author from NIST (2024) and International AI Safety Report (2026).

Figure 2 presents the logic of the study. It does not assume that more capable AI is automatically unsafe. Rather, capability growth can create or amplify risk pathways depending on the system, its users, and its deployment context. Safety controls should therefore be selected in response to a specific risk and evaluated for their remaining limitations.

6 Research Methodology

6.1 Research Design

This study will use a structured qualitative literature review. The review will synthesize research papers, technical reports, and international frameworks on future AI risks and practical safety measures. A literature based approach is appropriate because the study seeks to organize existing evidence, compare proposed safeguards, and identify gaps rather than test a single AI model or collect data from human participants.

6.2 Search Strategy and Sources

The search will be conducted through Google Scholar, IEEE Xplore, ACM Digital Library, arXiv, and the websites of credible institutions such as NIST, OECD, UNESCO, and the International AI Safety Report. Search phrases will include “future AI risks,” “AI safety,” “AI alignment,” “trustworthy AI,” “large language model safety,” “AI risk management,” “AI governance,” “AI reliability,” and “AI misuse.” Backward and forward citation searching will be used for highly relevant papers.

6.3 Eligibility Criteria

Criterion	Inclusion rule
Language	English-language sources.
Publication period	2020 to 2026, with older foundational work included where necessary.

Quality	Peer-reviewed papers, official technical reports, and recognized international frameworks.
Relevance	Direct discussion of AI risk, safety, reliability, alignment, governance, evaluation, or mitigation.
Exclusions	Duplicate records, unreferenced opinion pieces, and sources not directly related to AI safety or reliability.

6.4 Study Selection Procedure

Search results will first be screened by title and abstract. Full texts will then be reviewed against the eligibility criteria. A simple selection log will record the reason for excluding sources, for example duplication, insufficient credibility, or lack of direct relevance. The final set will deliberately include both peer-reviewed research and authoritative institutional reports so that technical evidence and practical governance guidance can be compared.

6.5 Data Extraction and Analysis

For each selected source, the study will record the publication details, AI risk category, proposed safeguard, evidence or rationale, application context, and stated limitation. The evidence will then be analysed using thematic analysis. Initial codes will include reliability, fairness, privacy, security, misuse, autonomy, monitoring, human oversight, and governance. Related codes will be grouped into themes, and the findings will be compared across technical, organizational, and governance responses.



Figure 3. Structured literature review process used in this study. Source: developed by the author.

6.6 Quality Assurance and Ethical Considerations

Source quality will be assessed by prioritizing peer-reviewed research and publications from recognized public institutions. Important claims will be checked against the original source rather than relying only on secondary summaries. The review will use the same extraction categories for every selected source and will retain a search log to make the process transparent. Claims about long-term risks will be described as uncertainties or scenarios unless supported by strong evidence; the proposal will distinguish observed current harms from possible future risks. The study uses public documents and collects no personal data from human participants. All sources will be cited accurately, and conclusions will avoid sensational or unsupported claims.

6.7 Research Tools and Evaluation Approach

The study will use Google Scholar, IEEE Xplore, ACM Digital Library, arXiv, and institutional repositories to identify sources. A spreadsheet or reference manager will be used to record search results,

citations, inclusion decisions, and extracted themes. No dataset, human participants, algorithm training, or software system will be used because the proposed design is a literature review. Instead of experimental accuracy metrics, the review will evaluate the strength of each source by its credibility, relevance, specificity of evidence, stated limitations, and applicability to safer and more reliable AI.

7 Expected Results and Outcomes

The study is expected to find that no single safeguard can make an AI system fully safe or reliable. Technical methods can reduce specific failure modes, but their effectiveness depends on context, model capability, data quality, and deployment practices. The expected contribution is a practical framework that links risks to a layered set of safeguards. This framework may help students, developers, and organizations make more informed decisions about designing, evaluating, and deploying AI systems.

Risk area	Illustrative safeguard	Important limitation
Unreliable outputs	Evaluation, retrieval, verification, human review	May still fail with weak sources or new contexts
Bias and unfairness	Dataset review, subgroup testing, impact assessment	Bias can persist through deployment and use
Security and misuse	Red-teaming, access controls, monitoring, incident response	Controls can be bypassed or reduce usefulness
Autonomy and oversight	Clear escalation rules, audit logs, human approval	Human oversight can become superficial at scale

8 Research Contribution and Significance

The study will contribute a clear, evidence-based mapping between AI risk categories and practical safeguards. Its academic contribution is to synthesize technical, organizational, and governance perspectives that are frequently examined separately. Its practical contribution is a usable framework that can help students and early-stage practitioners ask better questions before an AI system is deployed: what could fail, who could be harmed, which safeguard is appropriate, and what residual risk remains. The study will also make uncertainty visible rather than treating speculative long-term claims as established facts.

9 Scope and Limitations

This proposed study will not predict the exact future of AI, rank all risks by probability, or validate a safety technique through direct experiments. Its findings will depend on the quality and scope of the selected literature, and the fast-changing nature of AI means that some conclusions may need updating. These limitations are appropriate to a literature-review study and will be addressed through transparent source selection and careful distinction between evidence, interpretation, and uncertainty.

10 Ethical Considerations

This study will rely on public, properly attributed sources and will not collect personal data or involve human participants. It will represent risk claims proportionately, distinguish current documented harms from longer-term possibilities, and avoid presenting uncertain scenarios as established facts. The study will also recognize that safety measures can create trade-offs, such as restricting useful access or placing unequal

burdens on different users. Ethical analysis will therefore consider fairness, accountability, privacy, transparency, and human oversight alongside technical performance.

11 Research Timeline and Work Plan

The following is an indicative eight-week schedule for the full study. It can be refined after discussion with a prospective faculty supervisor and aligned with the selected Master's programme.

Stage	Main activity	Indicative timing
1. Preparation	Refine topic, research questions, search terms, and eligibility criteria.	Week 1
2. Evidence gathering	Search databases and institutional sources; screen titles, abstracts, and full texts.	Weeks 2 to 3
3. Analysis	Extract evidence, code themes, compare safeguards, and identify limitations.	Weeks 4 to 5
4. Writing and revision	Write findings, check citations, revise with supervisor feedback, and prepare final paper.	Weeks 6 to 8

12 Resources and Budget

The research is designed to use freely available or institutionally accessible resources. It has no expected participant, travel, laboratory, or specialized-computing costs. If a paid database, software license, or cloud service is required later, it will be requested through the department before use.

Resource	Purpose	Estimated cost
Academic databases and official reports	Evidence search and source verification	Nil
Computer and internet access	Searching, writing, and reference management	Existing
Reference manager or spreadsheet	Citation management and review matrix	Nil
Specialized hardware or participants	Not required for this literature review	Nil

13 Conclusion

Advanced AI systems can create real social and economic value, but their safety and reliability cannot be assumed from capability alone. This proposal sets out a structured literature review that will identify major risk categories, compare available safeguards, and develop a practical framework connecting risks with safeguards. Its value lies in treating AI safety as a layered responsibility: technical evaluation and monitoring must be supported by informed human oversight, organizational accountability, and governance. The proposal provides a focused, evidence-based starting point for Master's research on safer and more reliable AI systems and can be developed further with academic guidance in a suitable research group.

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